Hamza Mounir

Senior Software Engineer | Scalable Systems | Cloud Architecture | AI Systems | Technical Leadership

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Senior Software Engineer with extensive experience designing **scalable platforms, microservice architectures, and cloud-native systems**. I specialize in building reliable infrastructures that allow companies to scale from **early startup stages to enterprise-grade platforms**.

My expertise spans **backend architecture, distributed systems, DevOps automation, and AI-powered applications**. I build systems that are **resilient, automated, and operationally efficient**, reducing friction for both engineering teams and business operations.

On the frontend, I deliver **high-performance web applications** using React and Next.js, focusing on **user experience, accessibility, and seamless integration with complex backend services and AI-driven systems**.

I also design and deploy **AI-powered solutions**, including LLM-based services, retrieval-augmented generation (RAG), vector search, and context-aware systems. I build infrastructure to optimize token usage, manage streaming responses, and integrate intelligent automation into production environments.

Beyond coding, I actively improve **engineering processes, development workflows, and team collaboration**, mentoring engineers and ensuring teams can deliver faster with higher reliability.

Founder

Feen — AI Accounting Platform

https://www.feen.be

Feen is an **AI-powered accounting assistant for Belgian SMEs**, helping entrepreneurs manage invoices, transactions, and financial insights in a single platform.

The platform automates financial administration by combining document extraction, reconciliation, dashboards, and accounting collaboration tools.

Responsibilities

- Designed and built the full **SaaS architecture**
- Developed **AI-powered financial document extraction**
- Implemented **automatic transaction reconciliation**
- Built dashboards providing **real-time financial insights**
- Integrated **Peppol readiness and company verification tools**
- Developed scalable backend APIs and data pipelines
- Implemented LLM workflows for data processing and automation

Tech

Python • TypeScript • Next.js • AI/LLM APIs • Vector databases • Microservices • Cloud infrastructure

Synq — AI Agent Orchestration Platform

https://synq.pixldev.be

Synq is a **workflow orchestration platform for AI-native software delivery**, acting as the control plane between engineering teams and AI agents.

The platform transforms unstructured AI usage into **governed production pipelines**, enabling multiple AI agents to collaborate safely across development workflows.

Key features

- DAG-based workflow orchestration for AI-driven development pipelines

- Parallel AI agent execution (backend, frontend, QA, DevOps, security)
- Quality gates with automated testing and validation before deployment
- Sandboxed execution environments for safe experimentation
- Role-based permissions and governance for AI-generated code
- Cost intelligence tracking token usage and compute per agent
- Failure detection with automated recovery and rollback
- Plugin architecture supporting multiple AI providers (GPT, Claude, Codex, custom models)

Engineering

- Built orchestration engine coordinating **multi-agent AI workflows**
- Designed **event-driven pipeline system from prompt → production**
- Implemented streaming visibility into agent reasoning and execution
- Developed plugin system for integrating external tools and AI providers
- Built infrastructure for **AI cost tracking and optimization**

Technical Skills

Software Architecture & Backend Development

Languages
TypeScript • JavaScript • Python

Frameworks
NestJS • Node.js • gRPC • REST • GraphQL

Architecture
Microservices • Event-driven systems • Distributed systems • API-driven platforms

Scalability
High-availability systems • queue processing • service orchestration • financial-grade reliability

Devops
Azure • GCP • CI/CD • AKS / Docker

Ai engineering

- **Backend Developmen**t: Built and deployed tools leveraging context improvement, memory, and prompt engineering to optimize LLM calls and reduce token usage.

- **System Design**: Designed and implemented systems for streaming responses, model understanding, and iterative finetuning based on test feedback.
- **Infrastructure & Cloud**: Integrated vector databases and retrieval-augmented generation (RAG) patterns for enhanced information retrieval and semantic search.
- **Automation & Performance**: Automated deployment pipelines and ensured robust monitoring, error handling, and scalability for AI-powered services.

Frontend Development

- **Modern frameworks**: Proficient in React.js, Next.js, and TypeScript for building fast, scalable, and SEO-friendly applications in headless CMS environments
- **System Design, UI/UX**: Skilled in transforming UI/UX designs into pixel-perfect, responsive interfaces using TailwindCSS, and JavaScript and tools like Figma.
- **Performance & accessibility**: Focused on web performance optimization (lazy loading, SSR/ISR), and accessibility (WCAG, ARIA, semantic HTML) for inclusive, high-ranking web experiences.
- **Devops & collaboration**: Experienced with Git, Docker, and CI/CD workflows, ensuring robust development pipelines and effective cross-team collaboration.

Business-Critical Systems

- **FinTech & InsurTech Solutions**: Payments, pricing engines, compliance automation
- **Third-Party Integrations**: Banking APIs, insurance providers, external services
- **Contract & Policy Management**: Automated contract generation, bordereau reporting
- **Scalability & Optimization**: Data migration, performance tuning, distributed systems, dev tools, auto generation code

Leadership & Team Growth

- **Process Improvement**: Reducing inefficiencies in development workflows
- **Mentorship & Knowledge Sharing**: Supporting and guiding teams for better productivity
- **Cross-Team Collaboration**: Working with product, operations, and leadership

EXPÉRIENCE

Belfius, Brussels — AI Systems Engineer
July 2025 - Ongoing

At Belfius, I build AI systems that work within the strict constraints of a regulated bank. This means taking experimental AI technology and making it production-ready, secure, and compliant. I work closely with Architecture teams to ensure everything we build meets banking standards.

● Designed retrieval systems that let LLMs access bank knowledge safely, using vector databases and hybrid search
● Set up Azure infrastructure with proper security controls—managed identities, encryption, zero-trust policies.
● Created FastAPI services that connect AI models to enterprise data while maintaining strict access controls.
● Built AI agents that help with internal operations—automating tasks that previously required manual work.

My role focuses on bridging the gap between cutting-edge AI technology and enterprise-grade security requirements, ensuring every system is production-ready and meets regulatory standards.

Cohabs, Bruxelles — *Senior Software Engineer*
Avril 2025 - Juin 2025

At Cohabs, I focus on **building a scalable payments service to focus on all edges cases,** to help the team work more efficiently. And most important mirroring transactions to financial tool used by finance department, **accounting & payment consolidation.**

- **Third party api i**ntegration for communication to accounting tool.
- Enhance workflow on events with error managing and queue process.
- **Handle migration** between growing services and changes.
- Focused on **scalability, process automation, and system reliability** in a growing platform.

My role is not just about building software but **helping the team deliver with more reliability and tracing between departments.**

Greenomy, Bruxelles — *Senior Software Engineer*
Jan 2024 - Avril 2025

At Greenomy, I focus on **building a scalable SaaS platform while improving engineering workflows** to help the team work more efficiently.

- **Automated CI/CD processes**, improving deployment speed and reliability.
- Introduced **better development workflows, monitoring tools, and debugging strategies**.
- **Mentored engineers**, reducing friction in problem-solving and improving overall productivity.
- Create features based on business needs and auto scalable solutions.
- Focused on **scalability, process automation, and system reliability** in a growing platform.

My role is not just about building software but **helping the team deliver faster, more efficiently, and with better long-term stability**.

Qover, Bruxelles — *Senior Software Engineer*
2017 - 2024

I was part of the **early technical team at Qover**, helping transform it from a **13-person startup to a 150+ employee scale-up**. I worked across **all major systems**, ensuring that the platform was **scalable, automated, and built for long-term success**.

- Built **authentication, third-party integrations, pricing models, claims automation, and compliance systems**.
- Developed **whitelabel insurance solutions**, making it easy to onboard and launch new partners at scale.
- Created **contract automation, bordereau generation, financial reconciliation, and notification workflows**.
- Helped migrate and optimize cloud infrastructure from **AWS to GCP**, ensuring high availability.

- Played a key role in **building the foundation of a platform that is still running successfully at scale today**.

Qover was a **true end-to-end experience** in building a **high-growth Insurance as a service and embedded platform**, from **startup launch to full-scale enterprise operations**.

Emakina, Bruxelles — *Internship*
2017

- Developed a **digital signage platform** for real-time content display across multiple screens.
- Integrated **live data feeds (weather, schedules, news updates, etc.)**.
- Optimized **secure authentication and data access processes**.

FORMATION

Institut Paul Lambin, Bruxelles — *Bachelor in Computer Science*
2014 - 2017

COMPÉTENCES

Building Scalable, Self-Managed Platforms – Creating solutions that adapt and grow with the business.

Solving Real-World Business Problems – Aligning technical solutions with company objectives.

Improving Developer Efficiency – Reducing friction through automation, better tooling, and optimized workflows.

Mentoring and Team Leadership – Supporting engineers and promoting best practices.

LANGUES

French (Native)

English (Professional)